

Tracking Recovery Patterns in Acidified Lakes: A Paleolimnological Perspective

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Abstract

Long-term data are often lacking to effectively assess patterns of lake acidification and recovery. Fortunately, paleolimnological techniques can be used to infer past changes in lakewater acidity and related variables by means of biological indicators, such as diatom valves and chrysophyte scales, preserved in ²¹⁰Pb-dated sediment cores. We summarize paleolimnological data that we have gathered from 36 Sudbury (Ontario) and 20 Adirondack Park (New York) lakes to estimate the magnitude of lake acidification and any subsequent recovery in these lake systems. In both regions, many lakes were shown to have acidified considerably, some over two pH units, since the 1850s. Although some recovery was noted in both lake regions, Sudbury lakes generally showed larger increases in inferred lakewater pH with recent declines in sulfur emissions. Possible explanations of these differences include the greater decrease in sulfate deposition in the Sudbury area, as well as generally longer residence times of lakes in Sudbury, perhaps allowing for more in-lake alkalinity generation. In addition, Sudbury lakes generally had higher pre-industrial pH levels, suggesting that lakes with higher natural buffering capacities are more likely to recover more quickly

with declines in deposition, even if they had been acidified to a great extent.

Introduction

For several decades, lake acidification has been recognized as an important international problem affecting the water quality and biota of many lake systems. Research on the limnological effects of the atmospheric deposition of strong acids has been a major endeavour, especially in the 1980s, which has resulted in a large number of publications and reports (National Acid Precipitation Assessment Program 1990). These data frequently documented the adverse effects of acidification on lake ecosystems. Largely because of these studies, a number of legislative and voluntary measures were instigated which decreased significantly the amount of acidic deposition falling on many lake systems.

Although acidic deposition has declined in many lake regions, it is still unclear why some lakes seem to have responded positively to the decreased emissions, while others remain unresponsive. Part of the problem in dealing with these complex environmental issues is the lack of reliable long-term environmental monitoring data. A frequent complaint echoed by most environmental managers is that without such data it is often difficult to assess the magnitude of an environmental problem and, equally important, to prescribe any mitigation actions (Smol 1992, 1995). Similarly, it is difficult to evaluate whether mitigation efforts are having any desired effects without long-term monitoring data.

Fortunately, sediments often preserve a reliable archive of past limnological changes. Paleolimnological techniques have progressed to the point where statistically robust and ecologically sound approaches have been developed so that past lakewater conditions can be inferred with a high degree of reliability (Charles & Smol 1994). Paleolimnological studies of lake acidification have been used in a number of lake regions to determine if lakes have acidified and, if so, when and by how much (Charles et al. 1989; Battarbee et al. 1990; Charles & Smol 1990; Cumming et al. 1992a, 1994; Dixit et al. 1995).

We have published extensively on the use of paleolimnological techniques to infer patterns of lake acidification and recovery in the Sudbury region of Ontario (Dixit et al. 1995) and Adirondack Park of New York (Cumming et al. 1992a, 1994), as well as other regions (Charles et al. 1989). Collectively, these studies clearly indicate that different lakes follow different acidification trajectories, as well as different patterns of recovery. In some systems, no recovery was recorded, even though deposition had decreased.

In this overview paper we integrate some of these paleolimnological studies, published over a 12-year pe-

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riod, and attempt to provide some explanations as to why some lakes appear to recover from decreased acidic loading faster than other systems. Our analyses suggest that knowledge of pre-impact (pre-1850) acidity levels, as well as knowledge of when a lake acidified and under what levels of acidic deposition, are important in assessing potential trajectories of recovery from acidic deposition. This study provides another example of the importance of long-term paleolimnological data to the effective monitoring and management of lake systems.

Study Regions

The Sudbury region of Ontario (Fig. 1) is well known for the mining and smelting of copper-nickel ore, which began in the early 1880s (Keller 1992). It is a classic example of pollution from a point source. The emissions from Sudbury have affected thousands of lakes, with many of the lakes closest to the smelters becoming very acidic ($\text{pH} < 5$) during the periods of peak smelting activity. The region is also famous, however, for the dramatic decreases in emissions that have occurred since the 1970s, and the striking recovery in water chemistry patterns that have been witnessed in many of the local ecosystems. Most of the studies conducted in Sudbury and the surrounding region have been summarized in our various publications cited in this paper, as well as in a recent book (Gunn 1995).

The lakes in Adirondack Park are located in a forested, mountainous region of northern New York state (Fig. 1). In contrast to Sudbury, these lakes are distant

from any local point sources of major acidic emissions. Nonetheless, atmospheric deposition is still very acidic ($\text{pH} < 4.5$; Driscoll et al. 1991). Lakes in the Adirondack region have been extensively studied for the effects of recent lake acidification (Sullivan 1990; Driscoll et al. 1991). Past sulfate deposition in the Adirondacks has been estimated by Husar et al. (1991), who have shown that sulfate deposition increased in the latter part of the nineteenth century and then generally continued to increase further (with some declines in the 1930s and 1950s) until it peaked in 1970. Thereafter, sulfate deposition declined significantly, although the decrease in deposition is not as striking as the declines reported from the Sudbury region (Gunn 1995).

Methods

The paleolimnological methods we use are described in detail elsewhere (Cumming et al. 1994). Briefly, we remove lake sediment cores from a deep, flat basin near the center of a lake using a gravity corer specifically designed for high-resolution paleoenvironmental studies (Glew 1989, 1991). These sediments are sectioned at close intervals (between 0.25 and 1 cm, depending on the temporal interval being studied) with vertical extruders designed for high-resolution studies (Glew 1988), and the sediments are dated by ^{210}Pb chronology (for details, see Cumming et al. 1994).

Although we use many paleolimnological indicators to reconstruct lakewater conditions, our primary indicator groups for lake acidification studies in Sudbury and the Adirondacks are the siliceous diatom valves and scales of chrysophyte algae. For both these groups we have developed statistically robust and ecologically sound transfer functions whereby we can infer, with a high degree of reliability, past lakewater pH and related variables (Dixit et al. 1989a, 1991, 1993b; Cumming et al. 1992b).

Diatoms and/or chrysophyte assemblages are identified and enumerated in the various sediment sections. The assemblage data are then processed through the transfer functions, from which lakewater variables (e.g., pH) can be inferred (Charles & Smol 1994; Birks 1995). From these data we can reconstruct past environmental conditions and provide trajectories of past changes in lakewater acidity, from which the timing and rates of acidification and recovery can be assessed. By comparing our paleolimnological inference data to available long-term monitoring data, we have confirmed that these paleolimnological approaches are robust and defensible (Dixit et al. 1989b). Paleolimnological techniques therefore provide investigators with the ability to reconstruct past limnological conditions that were not measured in the past.

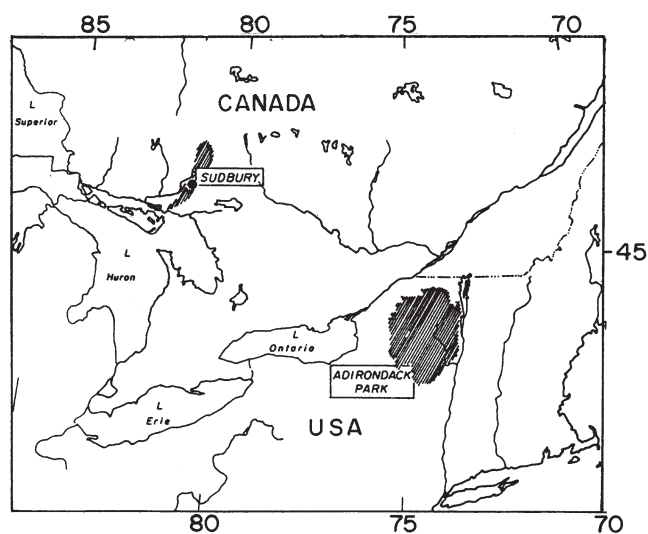


Figure 1. Map showing the Sudbury and Adirondack Park study areas. For details concerning the locations of individual study lakes, please see our original publications cited in the text.

Results and Discussion

Sudbury Lakes

Case Study: Baby Lake. In contrast to the data from the Adirondacks (see below), some Sudbury lakes have shown striking recovery patterns, many of which have been tracked by means of paleolimnological techniques. One case study (described in detail in Dixit et al. 1992a) is Baby Lake (46°24'N, 80°52'W), located about 1 km southwest of the Coniston Smelter, which operated from 1913 to 1972. Baby Lake is small (surface area 11.7 ha) but relatively deep, with a maximum depth of 25 m. The lake was monitored between 1968 and 1984 by Hutchinson and Havas (1986), who recorded a marked recovery in measured pH values in the years following the closure of the Coniston Smelter in 1972 (Fig. 2).

Paleolimnological data indicate striking changes over the last century in fossil diatom and chrysophyte assemblages preserved in a high-resolution, ^{210}Pb -dated sediment core from Baby Lake (described in detail by Dixit et al. 1992a). These floristic data were then used to infer past lakewater pH and related variables, using the transfer functions described by Dixit et al. (1989a, 1991, for chrysophytes and diatoms, respectively). The resulting pH inferences recorded the dramatic acidification of Baby Lake from a pH of about 6.5 in 1940 to a low of 4.2

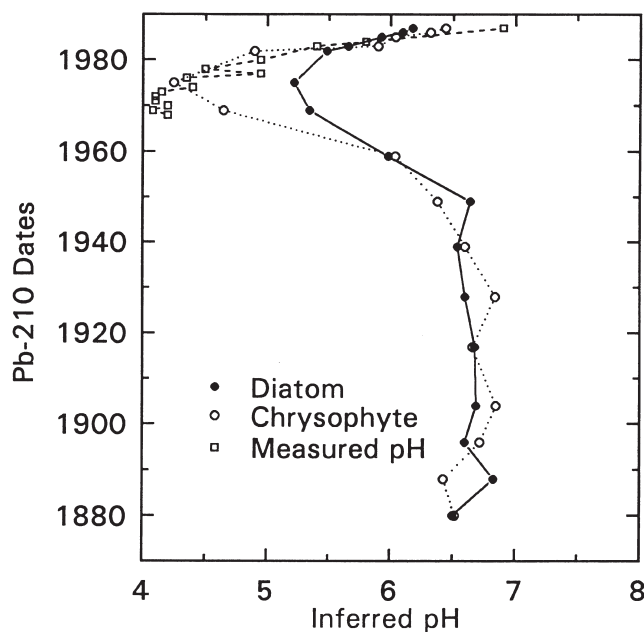


Figure 2. Chrysophyte-(open circles) and diatom-(closed circles) inferred pH changes from a 26-cm-long sediment core collected from Baby Lake, Sudbury. This analysis is described in detail in Dixit et al. (1992a). The measured pH data (marked by squares) were taken from Hutchinson and Havas (1986). This figure is modified from Dixit et al. (1992a).

in 1975. Following the closure of the Coniston smelter in 1972, the diatoms and chrysophytes recorded a striking recovery pattern in the lake. The pH trends inferred from diatom and chrysophyte assemblages (Fig. 2) faithfully tracked the recovery trends documented by Hutchinson and Havas (1986). Similar paleolimnological studies were conducted on a number of other Sudbury-area lakes (Dixit et al. 1989b, 1992c).

Regional Trends. In an attempt to synthesize some of the paleolimnological data that we have gathered on lake acidification and recovery in 36 Sudbury area lakes, we have summarized these results in a series of bar graphs (Fig. 3). Details for the individual lakes are provided primarily by Dixit et al. 1987, 1989b, 1990, 1991, 1992a, 1992b, 1992c, 1993a, 1995, 1996a, 1996b, 1996c. The paleolimnological data we use are based on the diatom and chrysophyte pH inferences from ^{210}Pb -dated sediment cores. We summarize these inferences in terms of (1) the maximum inferred acidification since approximately 1850 and (2) the change in inferred pH from approximately 1850 to the most recent sediment sample analyzed (Fig. 3). The difference between these two inferences represents any increases in inferred pH (i.e., recovery) that have occurred since the most acidic interval since about 1850. Some lakes did not have sufficient numbers of chrysophyte scales to allow us to reliably infer lakewater characteristics, and so fewer intervals are available for the chrysophyte data than for the diatom inferences (Fig. 3).

For example, the chrysophyte data indicate that Lumsden Lake acidified almost two pH units since the 1850s, but that pH has increased since the most acidic inferred value (Fig. 3). Meanwhile, chrysophyte inferences for Mountaintop Lake indicate a similar amount of acidification but no inferred recovery. All the Sudbury study lakes with a present pH of less than 6.6 have acidified since the 1850s. Some have acidified by over two pH units. As is typical for these types of data, chrysophytes tend to indicate larger changes in pH than diatoms do. The average inferred acidification (excluding lakes that were limed, see Fig. 3) was -0.6 pH units for the diatoms ($N = 32$) and -0.9 pH units for the chrysophytes ($N = 29$), perhaps reflecting the fact that chrysophytes typically bloom in the spring, when the lakes experience pH declines associated with acidic water from snowmelt (Cumming et al. 1992a; Dixit et al. 1992c). The diatom and chrysophyte data (Fig. 3) also indicate that approximately 40% of the lakes (limed lakes excluded) have increased in pH since the most acidic interval. For those lakes that showed pH increase, the average pH increases based on the chrysophyte and diatom data presented in Figure 3 (excluding the limed lakes) were 0.38 and 0.23 of a pH unit, respectively. Some of the lakes located on well-buffered catchments indicated increases in pH since the onset of acidic deposition. The reasons

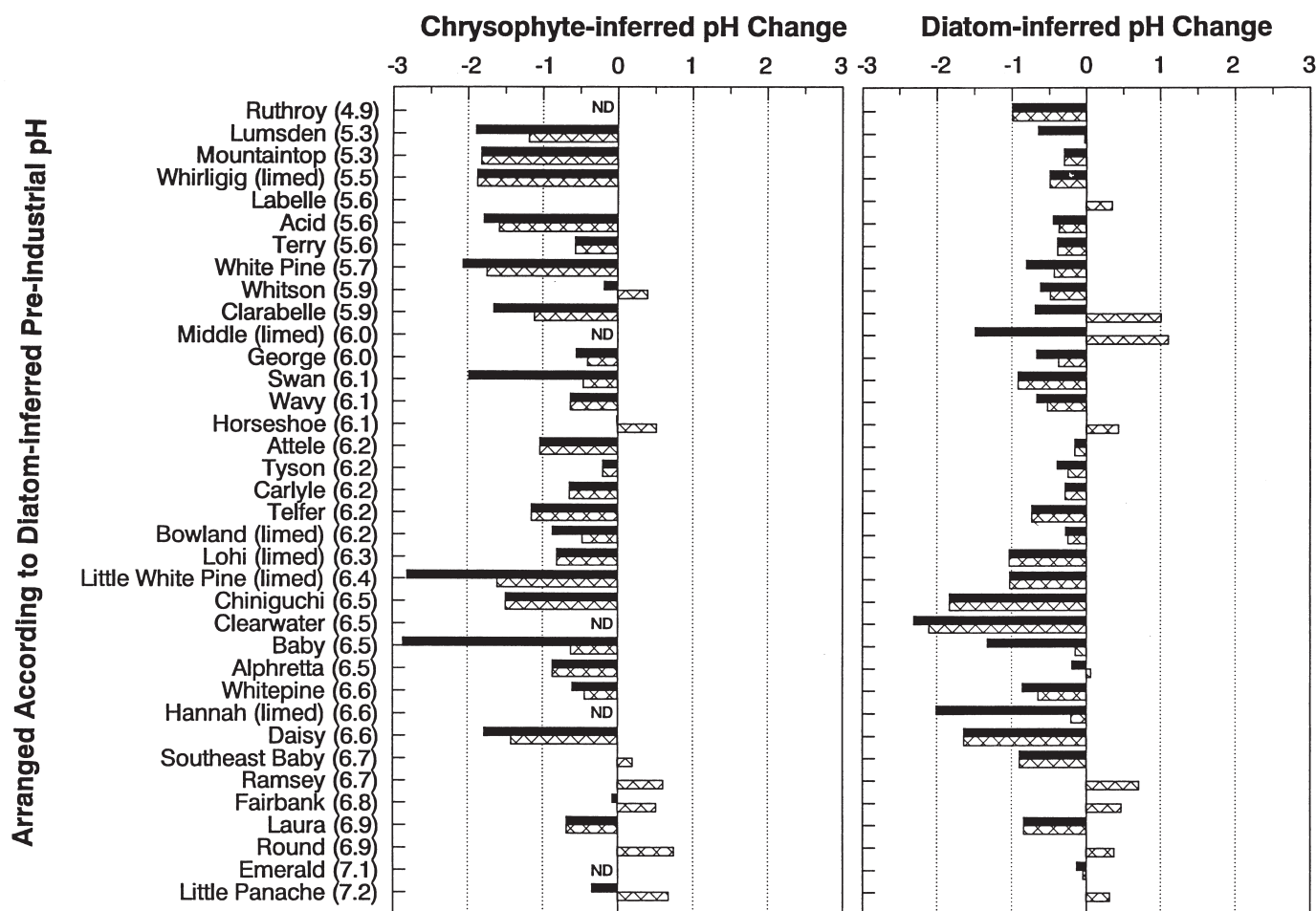


Figure 3. Regional trends in chrysophyte- and diatom-inferred lakewater pH change from 36 Sudbury area lakes. These paleolimnological data are based on the diatom and chrysophyte pH inferences from ^{210}Pb -dated sediment cores. Negative changes indicate a drop in pH. We summarize these inferences in terms of (1) the maximum inferred acidification since approximately 1850 (solid bars) and (2) the change in inferred pH from approximately 1850 to the most recent sediment sample analyzed (hatched bars). The difference between these two inferences represents any increases in inferred pH (i.e., recovery) that may have occurred since the most acidic interval since about 1850. A few lakes did not have sufficient numbers of chrysophyte scales to allow us to reliably infer lakewater characteristics; they are denoted by ND. The lakes are arranged in order of diatom-inferred pre-industrial pH (shown in brackets). Full details concerning the individual lakes are presented in the various papers by Dixit et al. cited in the text.

for these limnological changes are discussed elsewhere (Dixit et al. 1992b).

Adirondack Lakes

In a population-based paleolimnological study, Cumming et al. (1992a) showed that approximately 80% of Adirondack study lakes with present-day pH below 6.5 have acidified since pre-industrial times. Subsequently, Cumming et al. (1994) addressed the issue of recovery patterns in Adirondack Mountain lakes using paleolimnological approaches. This large project will be summarized briefly here.

Cumming et al. (1994) used the species assemblage changes of scaled chrysophytes in ^{210}Pb -dated sediment

cores from 20 low-alkalinity lakes in Adirondack Park to determine when these acid-sensitive lakes began to acidify and if they are still acidifying. Using cluster analysis of the chrysophyte-inferred pH profiles, they differentiated four categories of lake response to acidic deposition: (1) naturally well-buffered lakes that showed little or no evidence of acidification since pre-industrial times; (2) lakes with pre-industrial pH values (inferred from the fossil chrysophyte assemblages) between 5 and 6 that began to acidify around 1900; (3) "naturally" acidic lakes that acidified even further around 1900; and (4) lakes with pre-industrial pH values around 6 that acidified around 1930–1950.

Canonical variates analysis (CVA) in the above study clearly showed that background (or pre-impact) acidity

was the most important factor that could account for the four categories of lake response. For example, lakes that acidified first were those that naturally had lower pre-industrial pH values. But the above data also indicated that, despite considerable declines in acidic deposition since the 1970s (Husar et al. 1991), post-1970 trends in lakewater pH have been small and variable, suggesting that these low-alkalinity lakes have been relatively unresponsive to the post-1970 declines in sulfate deposition. An interesting finding of this survey was that some Adirondack lakes were naturally somewhat acidic, although they acidified further in this century.

The study by Cumming et al. (1994) used only chrysophyte microfossils, so no diatom data are presented here for the Adirondacks. An analysis similar to the one presented for Sudbury chrysophytes has been constructed for the 20 high-resolution Adirondack lake cores (Fig. 4) described by Cumming et al. (1994). As

with the Sudbury study lakes (Fig. 3), the chrysophyte inferences of pH indicate that these low-alkalinity lakes have acidified since 1850 (Fig. 4). The average arithmetic decline in pH since 1850 (excluding lakes that were limed, see Fig. 4) was similar to that for the Sudbury region (-0.96 pH units, $N = 15$). The Adirondack data also indicate that about 50% of the lakes have shown recent increases in pH, but these increases were much smaller (average pH increase 0.15) in the Adirondacks ($N = 15$) than those recorded in the Sudbury area (0.38 pH units, $N = 29$).

Comparison between the Sudbury and Adirondack Regions

The differences between the two study regions can be most effectively seen when the paleolimnological data are compared on the same graph (Fig. 5). Adirondack

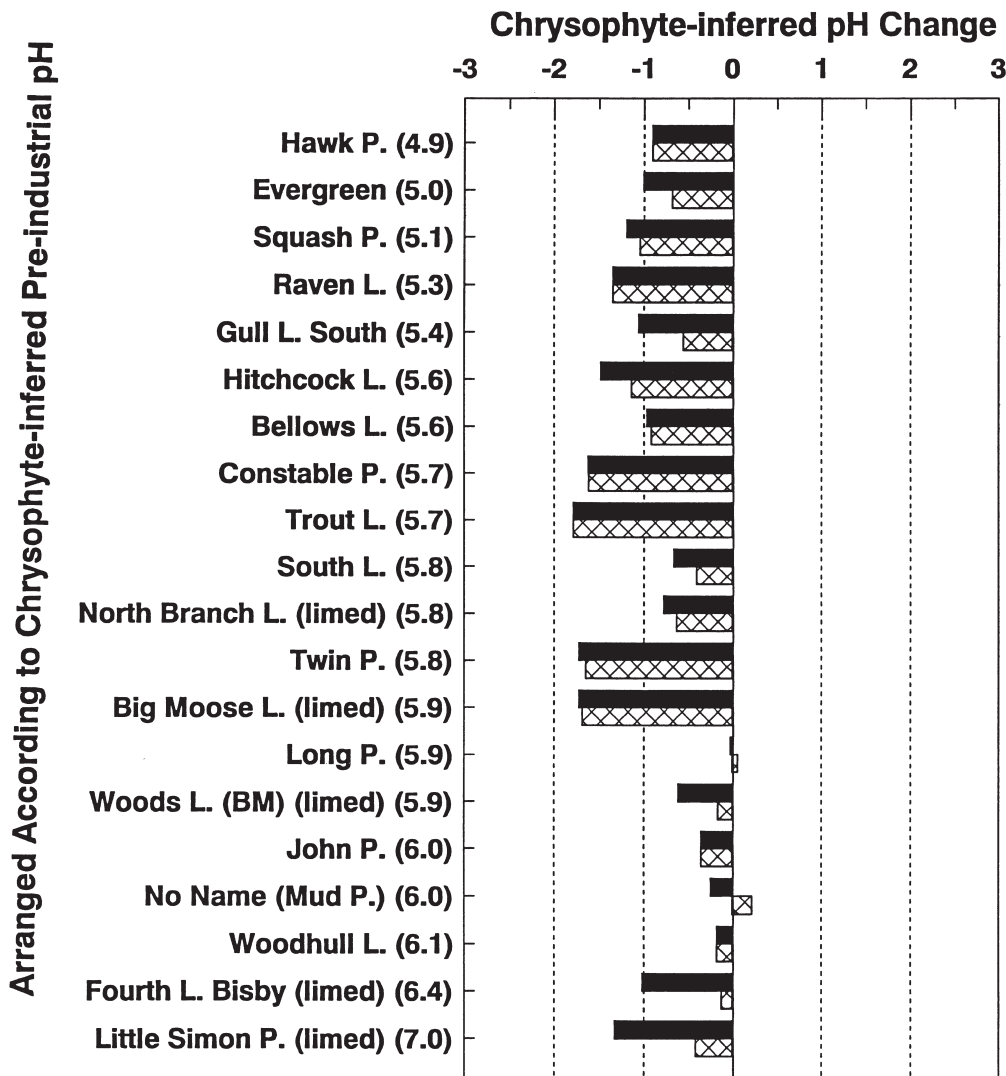


Figure 4. Regional trends in chrysophyte-inferred lake-water pH change from 20 Adirondack Park lakes. See the caption to Figure 3 for an explanation of the data presentation. The lakes are arranged in order of chrysophyte-inferred pre-industrial pH (shown in brackets). Diatom data are not available for most of these cores, so no diatom data are presented. Full details concerning the individual lakes are presented in Cumming et al. (1994).

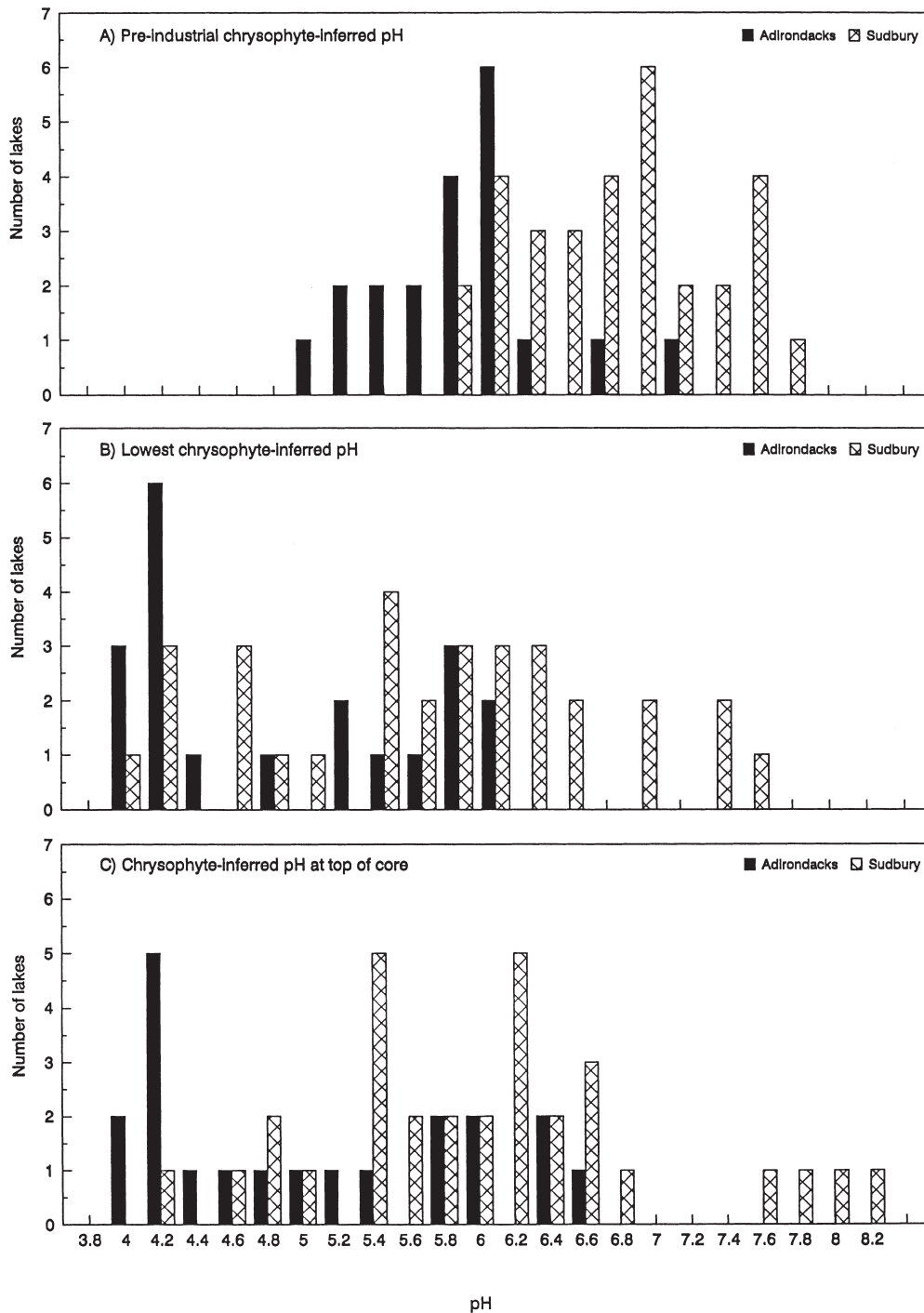


Figure 5. Comparison of frequency distributions between the Adirondack (solid bars) and Sudbury (hatched bars) study lakes for pre-industrial chrysophyte-inferred pH (a), lowest chrysophyte-inferred pH for the core (b), and chrysophyte-inferred pH at the top of each core (recent sediments) (c). Frequency distributions were calculated based on 0.2 pH increments from chrysophyte-inferred pH values from 4.0 to 8.2.

lakes clearly had lower pre-industrial pH values than those inferred in the Sudbury study lakes (Fig. 5a). However, a large number of lakes acidified considerably in both lake regions, as is documented by the marked shift in the number of lakes according to their lowest chrysophyte-inferred pH (Fig. 5b). When we examined the shift in the distribution of lakes in both regions based on pH inferences from the lakes' recent

sediments, recovery was more evident in the Sudbury region than in the Adirondacks (Fig. 5c).

Synthesis and Conclusions

Our paleolimnological studies indicate marked acidification in a large suite of acid-sensitive lakes in both Sudbury and Adirondack Park. In the Adirondack data

set, lakes that were somewhat naturally acidic (pH 5–6) were the first to acidify and showed little evidence of recovery with recent declines in sulfate deposition. In the Sudbury lake data set, we also recorded marked pH declines, especially in lakes that are located in close proximity to the smelters (e.g., Swan, Clearwater, Baby, and Daisy lakes). Interestingly, some of the Sudbury lakes with relatively high background pH values (e.g., Daisy and Baby lakes) still acidified greatly, despite their relatively high buffering capacities. This is a testament to the amount of acidic deposition that these lakes received. Nonetheless, many experienced striking recovery patterns (Figs. 2 & 3).

Our data sets are not sufficiently large to determine unequivocally why some lakes have recovered while others appear to have undergone little change. But some tentative general observations can be made.

Lakes in the Sudbury region tended to show more recovery than lakes in Adirondack Park. As noted above, Sudbury lakes tended to have higher pre-industrial pH values and had naturally higher buffering capacities. Undoubtedly, an important factor has also been that the recent declines in acidic deposition have been strikingly different between the two regions. Husar et al. (1991) estimated that sulfate deposition on Adirondack lakes has declined about 30% since the 1970s. Unfortunately, except for relatively recent measurements, we do not know how high past sulfate deposition was in Sudbury, but it was definitely much higher than in the Adirondack region, with both large wet and large dry components (Dillon et al. 1982). In the 1980s, study lakes within 100 km of Sudbury had average annual sulfate concentrations ranging from 220 to 265 $\mu\text{eq/L}$ (Keller et al. 1992), and this was after sulfate had already shown large declines since the 1970s, when sulfate concentrations up to 400–500 $\mu\text{eq/L}$ were recorded in some lakes (Keller & Pitblado 1986). In contrast, most Adirondack lakes had sulfate concentrations in the range of 50–200 $\mu\text{eq/L}$ during 1984–1987 (Kretser et al. 1989; Driscoll et al. 1991). Thus, historical sulfur deposition was clearly much higher in Sudbury.

Decreases in lake sulfate levels have also been much more dramatic in Sudbury. During the 1980s, lake-water sulfate declined about 3 $\mu\text{eq/L/yr}$, on average, in the Sudbury study lakes (Keller et al. 1992), while sulfate in Adirondack lakes declined, on average, only 1.8 $\mu\text{eq/L/yr}$ from 1982–1994 (Driscoll et al. 1995). Moreover, large lake sulfate declines had already occurred around Sudbury between the mid 1970s and the early 1980s (Keller & Pitblado 1986). Unfortunately, total acidic deposition in these two regions has not been well documented. However, the relative differences in past sulfate loading and changes in deposition are consistent with our paleolimnological data. Additional factors complicating recovery include increased nitrogen leaching from Ad-

irondack watersheds, and decreases in the atmospheric deposition of base cations, which may be offsetting the benefits of sulfate reductions on lake acidity (Driscoll & Van Dreason 1993).

A further complicating factor in assessing trends between Sudbury and the Adirondacks is differences in hydrology. Although hydrologic data are not available for many of the lakes used in our analyses, water renewal times are typically measured on the order of years for the Sudbury lakes (W. Keller, personal communication), while flushing rates in our Adirondack lakes are typically measured in months (Cumming et al. 1994). Within-lake alkalinity generation from sulfate and nitrate reduction would be expected to be greater in lakes with longer renewal times (Schindler 1986).

This study shows that even if monitoring data are not available (e.g., pH readings over the last 150 years), these data can still be inferred using paleolimnological techniques. As noted by Smol (1992), Sörenson only established the pH convention in 1909, yet for many management questions it is important to know what limnological conditions were like in the 1800s. As exemplified in this paper, by knowing the pre-industrial pH levels in a suite of study lakes and by documenting the changes in lake-water chemistry that occurred as a result of industrialization, we can better understand the stresses that human activities have placed on lake ecosystems. Paleolimnological techniques are now available to assess many different types of environmental degradation and recovery in addition to acidification (Smol 1995).

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LITERATURE CITED

- Battarbee, R. W., J. Mason, I. Renberg, and J. F. Taillandier. 1990. *Paleolimnology and lake acidification*. The Royal Society, London.
- Birks, H. J. B. 1995. Quantitative palaeoenvironmental reconstructions. Pages 161–254 in D. Maddy and J. S. Brew, editors. *Statistical modelling in Quaternary science data*. Technical guide 5. Quaternary Research Association, Cambridge, United Kingdom.

- Charles, D. F., and J. P. Smol. 1990. The PIRLA II project: regional assessment of lake acidification trends. *Verhandlungen der Internationalen Vereinigung von Limnologen* **24**:474–480.
- Charles, D. F., and J. P. Smol. 1994. Long-term chemical changes in lakes: quantitative inferences using biotic remains in the sediment record. Pages 3–31 in L. Baker, editor. *Environmental chemistry of lakes and reservoirs. Advances in Chemistry Series 237*. American Chemical Society, Washington, D.C.
- Charles, D. F., R. W. Battarbee, I. Renberg, H. van Dam, and J. P. Smol. 1989. Paleoecological analysis of lake acidification trends in North America and Europe using diatoms and chrysophytes. Pages 207–276 in S. A. Norton, S. E. Lindberg, and A. L. Page, editors. *Acid precipitation. Volume 2*. Springer-Verlag, Stuttgart, Germany.
- Cumming, B. F., J. P. Smol, J. C. Kingston, D. F. Charles, H. J. B. Birks, K. E. Camburn, S. S. Dixit, A. J. Uutala, and A. R. Selle. 1992a. How much acidification has occurred in Adirondack region (New York, USA) lakes since pre-industrial times? *Canadian Journal of Fisheries and Aquatic Sciences* **49**:128–141.
- Cumming, B. F., J. P. Smol, and H. J. B. Birks. 1992b. Scaled chrysophytes (Chrysophyceae and Synurophyceae) from Adirondack (N.Y., USA) drainage lakes and their relationship to environmental variables. *Journal of Phycology* **28**:162–178.
- Cumming, B. F., K. Davey, J. P. Smol, and H. J. B. Birks. 1994. When did acid-sensitive Adirondack lakes (New York, USA) begin to acidify and are they still acidifying? *Canadian Journal of Fisheries and Aquatic Sciences* **51**:1550–1568.
- Dillon, P. J., D. S. Jeffries, and W. A. Scheider. 1982. The use of calibrated lakes and watersheds for estimating atmospheric deposition near a large point source. *Water, Air and Soil Pollution* **18**:241–258.
- Dixit, S. S., A. S. Dixit, and R. D. Evans. 1987. Paleolimnological evidence of recent acidification in two Sudbury (Canada) lakes. *The Science of the Total Environment* **67**:53–67.
- Dixit, S. S., A. S. Dixit, and J. P. Smol. 1989a. Relationship between chrysophyte assemblages and environmental variables in 72 Sudbury lakes as examined by Canonical Correspondence Analysis (CCA). *Canadian Journal of Fisheries and Aquatic Sciences* **46**:1667–1676.
- Dixit, S. S., A. S. Dixit, and J. P. Smol. 1989b. Lake acidification recovery can be monitored using chrysophyte microfossils. *Canadian Journal of Fisheries and Aquatic Sciences* **46**:1309–1312.
- Dixit, S. S., A. S. Dixit, and J. P. Smol. 1990. Paleolimnological investigation of three manipulated lakes from Sudbury, Canada. *Hydrobiologia* **214**:245–252.
- Dixit, S. S., A. S. Dixit, and J. P. Smol. 1991. Multivariable environmental inferences based on diatom assemblages from Sudbury (Canada) lakes. *Freshwater Biology* **26**:251–265.
- Dixit, A. S., S. S. Dixit, and J. P. Smol. 1992a. Algal microfossils provide high temporal resolution of environmental trends. *Water, Air and Soil Pollution* **62**:75–87.
- Dixit, S. S., A. S. Dixit, and J. P. Smol. 1992b. Assessment of changes in lake water chemistry in Sudbury area lakes since preindustrial times. *Canadian Journal of Fisheries and Aquatic Sciences* **49**(Suppl. 1):8–16.
- Dixit, A. S., S. S. Dixit, and J. P. Smol. 1992c. Long-term trends in lake water pH and metal concentrations inferred from diatoms and chrysophytes in three lakes near Sudbury, Ontario. *Canadian Journal of Fisheries and Aquatic Sciences* **49**(Suppl. 1):17–24.
- Dixit, A. S., S. S. Dixit, and J. P. Smol. 1993a. Post-industrial acidification and metal contamination in Whitepine Lake (Sudbury, Canada): a paleolimnological perspective. *Journal of Paleolimnology* **9**:141–146.
- Dixit, S. S., B. F. Cumming, J. C. Kingston, J. P. Smol, H. J. B. Birks, A. J. Uutala, D. F. Charles, and K. Camburn. 1993b. Diatom assemblages from Adirondack lakes (N.Y., USA) and the development of inference models for retrospective environmental assessment. *Journal of Paleolimnology* **8**:27–47.
- Dixit, S. S., A. S. Dixit, J. P. Smol, and W. Keller. 1995. Reading the records stored in the lake sediments: a method of examining the history and extent of industrial damage to lakes. Pages 33–44 in J. Gunn, editor. *Restoration and recovery of an industrial region*. Springer-Verlag, New York.
- Dixit, A. S., S. S. Dixit, and J. P. Smol. 1996a. Long-term trends in limnological characteristics in the Aurora Trout Lakes, Sudbury, Canada. *Hydrobiologia* **335**:171–181.
- Dixit, A. S., S. S. Dixit, and J. P. Smol. 1996b. Long-term water quality changes in Ramsey Lake (Sudbury, Canada) as revealed through paleolimnology. *Journal of Environmental Science and Health* **A31**:941–956.
- Dixit, A. S., S. S. Dixit, and J. P. Smol. 1996c. Setting restoration goals for an acid metal-contaminated lake: a paleolimnological study of Daisy Lake (Sudbury, Canada). *Lake and Reservoir Management* **12**:323–330.
- Driscoll, C. T., and R. Van Dreason. 1993. Seasonal and long-term temporal patterns in the chemistry of Adirondack lakes. *Water, Air and Soil Pollution* **67**:301–317.
- Driscoll, C. T., R. M. Newton, C. P. Gubala, J. P. Baker, and S. W. Christensen. 1991. Adirondack Mountains. Pages 133–202 in D. F. Charles, editor. *Acidic deposition and aquatic ecosystems: regional case studies*. Springer-Verlag, New York.
- Driscoll, C. T., K. M. Postek, W. Kretser, and D. J. Raynal. 1995. Long-term trends in the chemistry of precipitation and lake water in the Adirondack region of New York, USA. *Water, Air and Soil Pollution* **85**:583–588.
- Glew, J. R. 1988. A portable extruding device for close interval sectioning of unconsolidated core samples. *Journal of Paleolimnology* **1**:235–239.
- Glew, J. R. 1989. A new trigger mechanism for sediment samplers. *Journal of Paleolimnology* **2**:241–243.
- Glew, J. R. 1991. Miniature gravity corer for recovering short cores. *Journal of Paleolimnology* **5**:285–287.
- Gunn, J. M., editor. 1995. *Restoration and recovery of an industrial region: progress in restoring the smelter-damaged landscape near Sudbury, Canada*. Springer-Verlag, New York.
- Husar, R. B., T. J. Sullivan, and D. F. Charles. 1991. Methods for assessing long-term trends in atmospheric deposition and surface water chemistry. Pages 65–82 in D. F. Charles, editor. *Acidic deposition and aquatic ecosystems: regional case studies*. Springer-Verlag, New York.
- Hutchinson, T. C., and M. Havas. 1986. Recovery of previously acidified lakes near Coniston, Canada, following reductions in atmospheric sulphur and metal emissions. *Water, Air and Soil Pollution* **29**:319–333.
- Keller, W. 1992. Introduction and overview to aquatic acidification studies in the Sudbury, Ontario, Canada, area. *Canadian Journal of Fisheries and Aquatic Sciences* **49**(Suppl. 1):3–7.
- Keller, W., and J. R. Pitblado. 1986. Water quality changes in Sudbury area lakes: a comparison of synoptic surveys in 1974–76 and 1981–83. *Water, Air and Soil Pollution* **29**:285–296.
- Keller, W., J. R. Pitblado, and J. Carbone. 1992. Chemical responses of acidic lakes in the Sudbury, Ontario, area to reduced smelter emissions, 1981–1989. *Canadian Journal of Fisheries and Aquatic Sciences* **49**(Suppl. 1):25–32.
- Kretser, W., J. Gallagher, and J. Nicolette. 1989. Adirondack Lakes Study 1984–1987: an evaluation of fish communities and water chemistry. Report to the Adirondack Lakes Survey Corporation, Ray Brook, New York.
- National Acid Precipitation and Assessment Program. 1990. *Integrated assessment*. National Acid Precipitation and Assess-

- ment Program, Washington, D.C.
- Schindler, D. W. 1986. The significance of in-lake production of alkalinity. *Water, Air and Soil Pollution* **30**:931–944.
- Smol, J. P. 1992. Paleolimnology: an important tool for effective ecosystem management. *Journal of Aquatic Ecosystem Health* **1**:49–58.
- Smol, J. P. 1995. Paleolimnological approaches to the evaluation and monitoring of ecosystem health: providing a history for environmental damage and recovery. Pages 301–318 in D. Rapport, C. Gaudet, and P. Calow, editors. *Evaluating and monitoring the health of large scale ecosystems*. NATO ASI Series, Volume **128**. Springer-Verlag, Stuttgart.
- Sullivan, T. J. 1990. Historical changes in surface water acid-base chemistry in response to acidic deposition. Report 11. Pages 11-1–11-181 in *State of science and technology*. National Acid Precipitation Assessment Program, Washington, D.C.